

DATA SCIENCE PROJECT

Bike Sharing Demand Prediction

Objective: Predict the hourly number of bike rentals using temporal, calendar, and weather-related features

 **Presentation:** Heike Thöricht

 **Bremen**, 23. August 2026

Agenda



The Challenge

Why demand forecasting matters



Our Approach

Data-driven solution



Model Performance

Proven accuracy



Key Insights

What drives demand

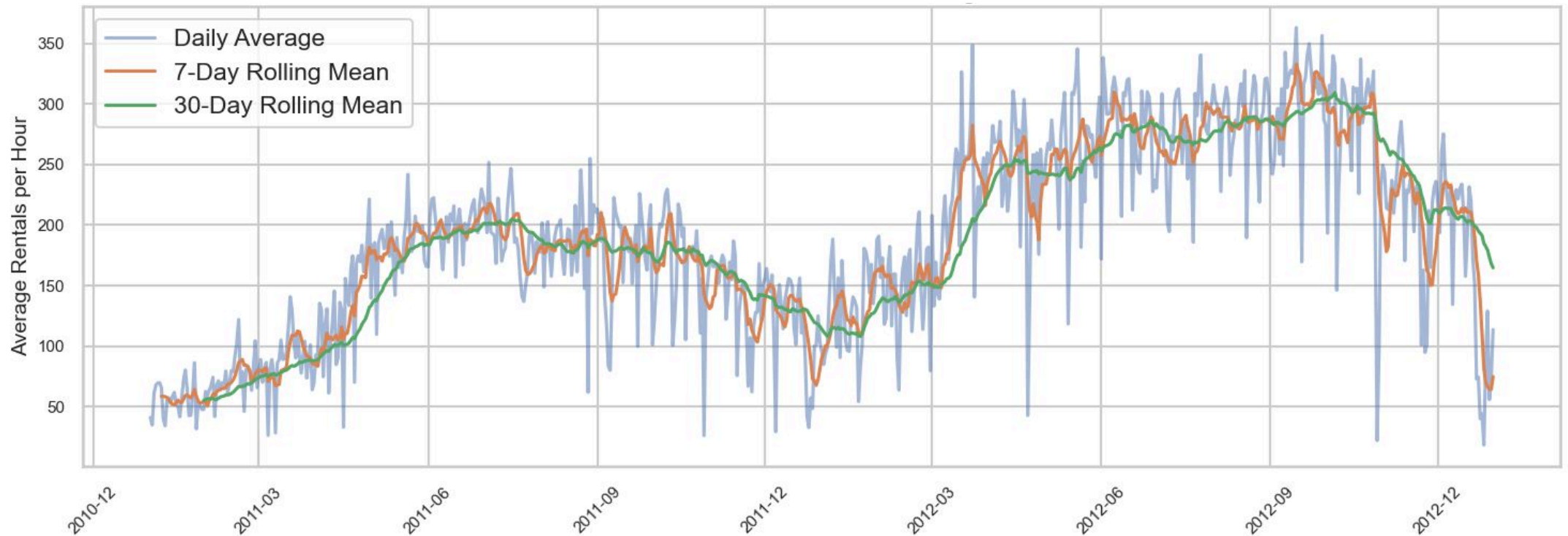


Implementation & Next Steps

How to use it

Demand Varies Over Time

The observed trend and seasonal variation make chronological validation essential for realistic forecast evaluation.



□ Because future demand cannot be used to predict the past, models were evaluated with time-series cross-validation.

Executive Summary: A Validated Forecasting Prototype



Prediction Accuracy

89.1% of demand variance explained ($R^2 = 0.891$) — strong, validated accuracy for reliable planning



Forecast Error

Average forecast error: 33 bikes per hour (MAE = 33.0) — realistic margin confirmed through time-series cross validation



Key Driver

Historical demand patterns (lag_1: 30.7%) and time-of-day (14.1%) drive predictions — highly actionable

The model demonstrates strong performance in time-series cross-validation and provides a validated prototype for a controlled operational pilot.

Evaluation approach: Time-series cross-validation to preserve chronological order and reflect the forecasting scenario

Our Data Foundation

Real-world hourly rental data from the Capital Bikeshare system, providing a relevant operational context for demand forecasting.

- **17,379** hourly observations from real bike-sharing operations
- **2 years** of historical data (2011–2012)
- Complete data quality — no missing values

The Business Problem

Without reliable demand forecasts, your planning division faces avoidable inefficiencies and operational risk.

- Unpredictable demand creates **logistics and staffing challenges**
- Core question: **How many bikes are needed each hour?**
- Key insight: **Historical demand patterns (30.7%) + time-of-day (14.1%)** are the strongest predictors
- Goal: Accurate hourly forecasts based on **demand momentum and temporal patterns**

Key Insight: Demand Patterns Drive Planning



Commuter Peaks

Working days show demand spikes at **8 AM** and **5–6 PM** — align bike availability with commute peaks.



Leisure Patterns

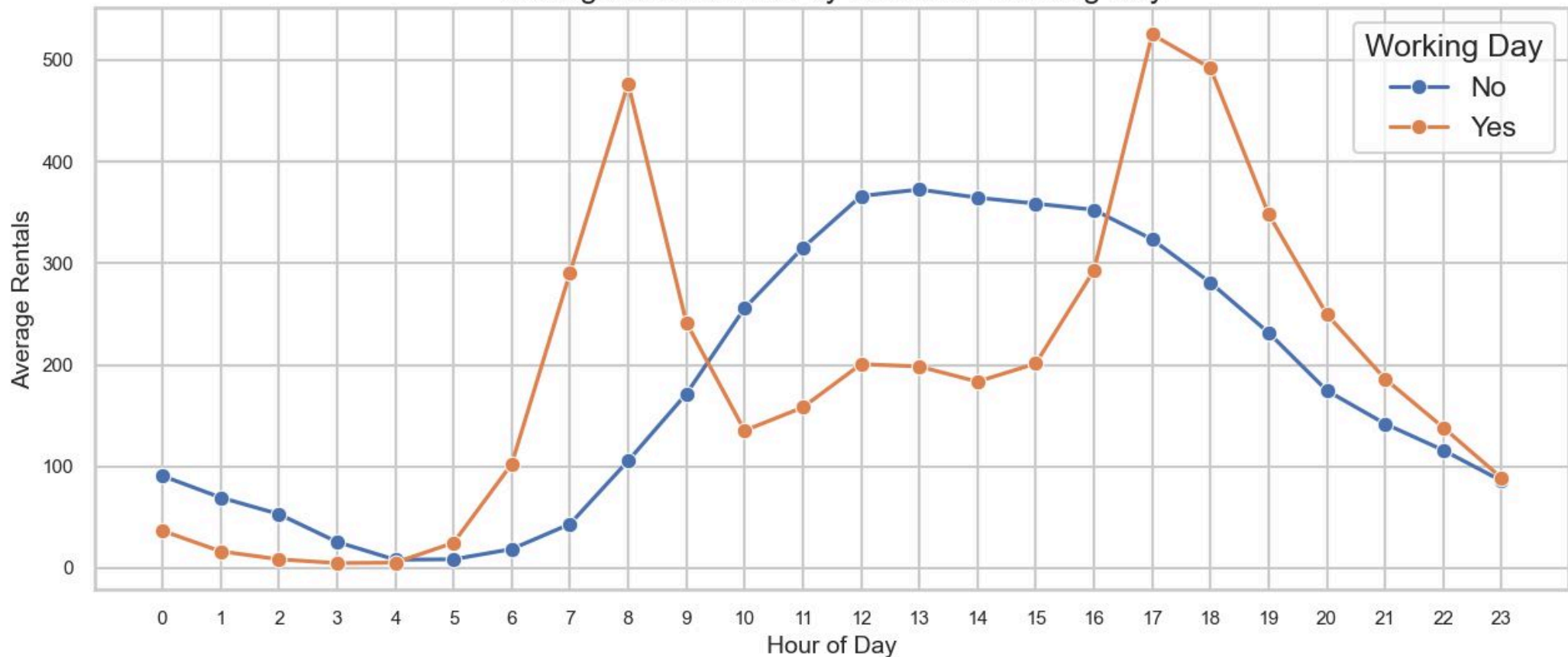
Non-working days show midday peaks — different staffing and bike distribution needed



Actionable

Hour-of-day is the **key planning signal** — enables precise hourly planning

Average Bike Rentals by Hour and Working Day



These recurring patterns provide useful signals for hourly planning.

Model Performance: Validated Forecast Accuracy

XGBoost achieved the strongest forecast performance in time-series cross-validation, with $R^2 = 0.891$, MAE = 33.0 rentals per hour and RMSE = 50.6.

Model	Feature Set	R^2	MAE	RMSE
Linear Regression	V01	0.876	39.7 bikes	56.5
Random Forest	V01	0.861	36.2 bikes	57.2
 XGBoost (Optuna CV)	V01	0.891	33.0 bikes	50.6

Winner: XGBoost

XGBoost achieved the strongest cross-validated forecast performance among the tested models.

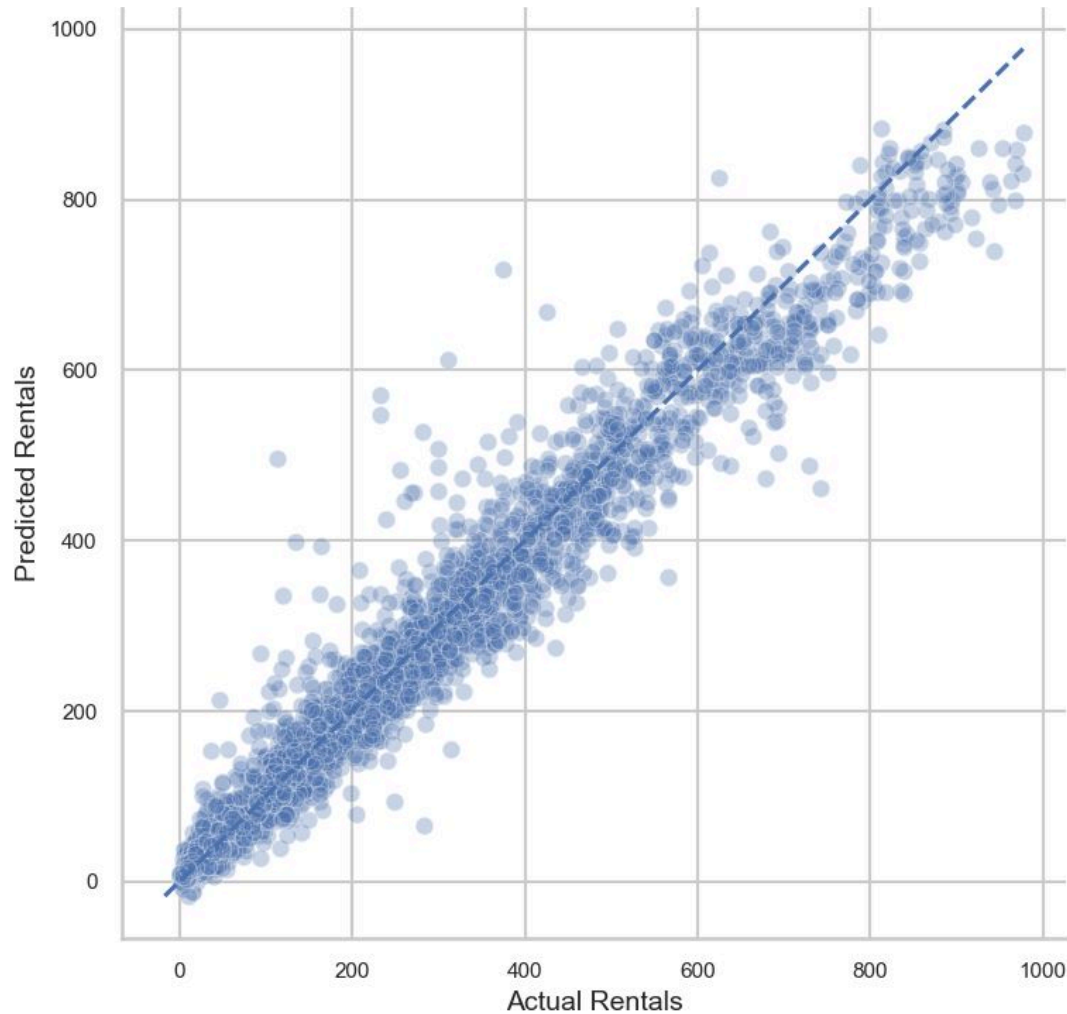
Forecast error: 33 rentals per hour

The MAE provides a transparent benchmark for evaluating suitability in a controlled pilot.

Ready for pilot evaluation

Strong time-series validation supports a controlled pilot; live-data integration and ongoing monitoring would be required before deployment.

Validation: Performance and Pilot Considerations



✓ Overall performance

Predictions closely track actual demand across the evaluation period.

⚠ Peak-hour risk

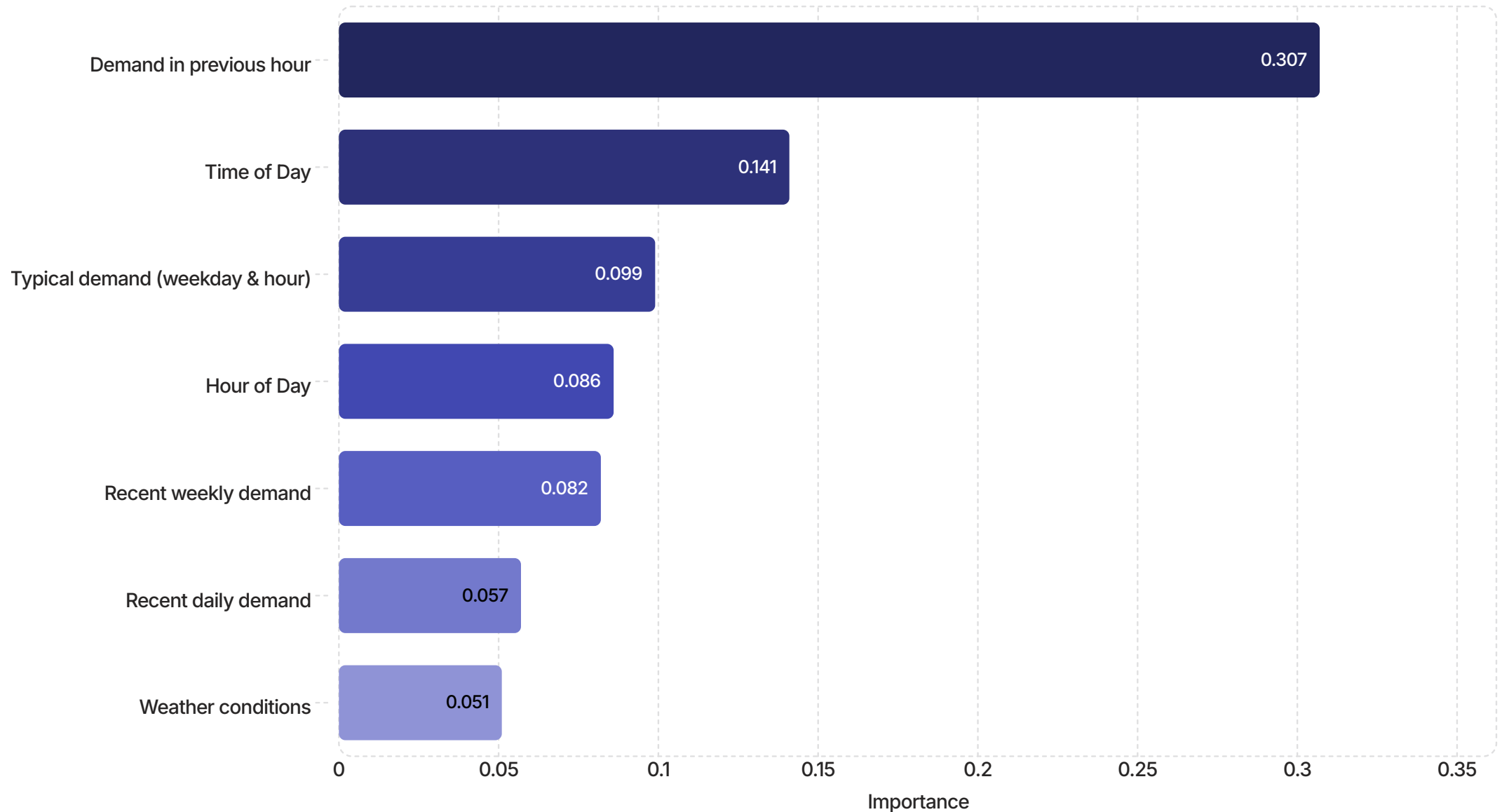
Forecast deviations are larger during demand spikes; peak periods require operational buffer capacity

🎯 Potential for operational use

Evaluate forecasts in daily planning during a controlled pilot, supported by weekly error monitoring and a predefined threshold such as $MAE < 50$ rentals.

What Drives Demand: The Key Factors

The model's top predictors reveal a clear pattern: recent behavior and time of day dominate, with weather as a fine-tuning signal.



Recent demand is the strongest individual predictor, with a feature-importance value of 30.7%.

How the Pilot Works

A simple operating model for next-hour demand forecasts.



Refresh the Forecast

EVERY HOUR

Update the next-hour forecast using the latest rental data.



Use It in Operations

EVERY DAY

Support bike availability, rebalancing and staffing decisions.



Monitor and Improve

EVERY WEEK

Review overall forecast error and peak-hour performance.



Model Updates

Retrain quarterly or earlier if forecast quality declines.



Pilot Success Target

Average forecast error below **50 bikes per hour**.

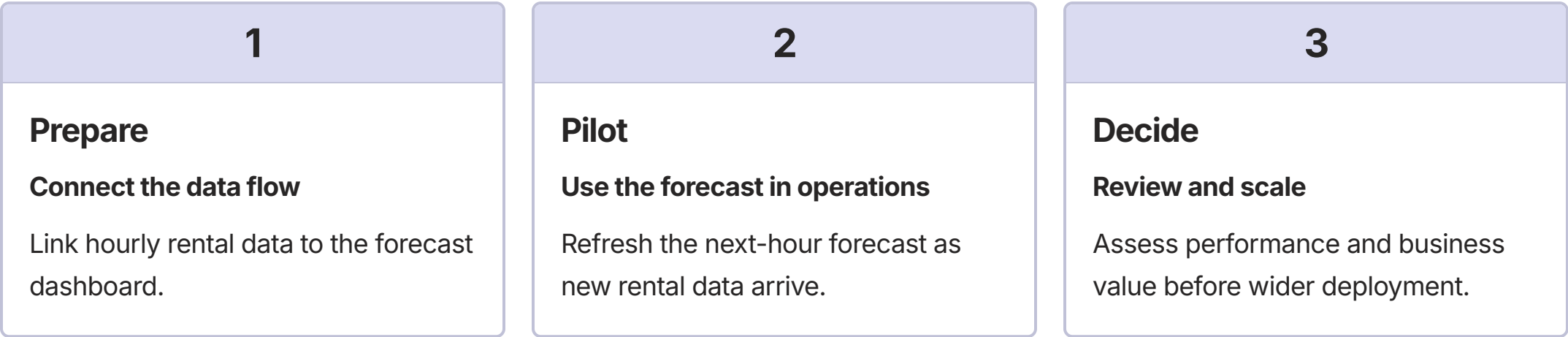


Owners

Operations · Data & Analytics · IT

Next Step: Launch a Controlled Pilot

Recommendation: Start a 4-week pilot using hourly forecasts to support bike availability, rebalancing and staffing decisions.



Pilot success criteria

- Average forecast error remains below **50 bikes per hour**
- Peak-hour performance is monitored and buffers are defined
- Operations confirms that the forecast improves planning decisions

Decision today: Agree pilot scope, nominate owners and confirm the start date.

Week 1: Dashboard and data review · Week 2: Technical integration check · Weeks 3–6: Controlled pilot

 BACKUP SLIDE

Appendix: How Key Factors Influence the Forecast

SHAP Summary Plot

SHAP analysis confirms that recent demand and typical weekday-hour patterns drive the forecast, while time-of-day and weather provide smaller refinements.

